

Übungsblatt 0a (Repetition Terme)

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Ausmultiplizieren

1. Aufgabe

Vereinfache die folgenden Terme indem du die Klammern auflöst:

(a)

$$(3x + 2)(2x - 3)(x - 1) - (x - 1)(x + 2)(x - 3) = ?$$

(b)

$$(3x + 2)(5x + 7) - \{-[-3(5x - 3)(7x + 5) - (2x + 8)(3x + 7)]\} = ?$$

(c)

$$-a - (-b) + \{-3a - [-7 + (2a - 3b)]\} = ?$$

(d)

$$(3x - 7)^2 - (3x + 1)(3x - 1) = ?$$

(e)

$$(1 - a^3)^3 + (a^3 - 1)^3 = ?$$

(f)

$$(1 - a)^2 - ((a - 1)^2 - (a^2 - 1)) = ?$$

(g)

$$a^2b^2 - (a^2 - 1)(b^2 + 1) - (a + b)^2 = ?$$

(h)

$$-((2 - a)(3 - b) - ((4 - c)(5 - d) - (a + c)(b - d))) = ?$$

Faktorisieren

2. Aufgabe

Faktorisiere soweit wie möglich:

(a)

$$1 - 2a + a^2 - b^2 = ?$$

(b)

$$6a + 3ac - 2b - bc = ?$$

(c)

$$(x - 2)(2a + 3b) - (a - b)(x - 2) + (b + a)(2 - x) = ?$$

(d)

$$x - y - x^2 + 2xy - y^2 = ?$$

(e)

$$u^2 - 14u + 49 = ?$$

(f)

$$12 - 8z + z^2 = ?$$

(g)

$$16 - x^2 + 4xz^2 - 4z^4 = ?$$

(h)

$$(a - b)(2x + 3y) - (2x - y)(a - b) - (b - a)(x - 3y) = ?$$

(i)

$$xy - x - yz - y + z + 1 = ?$$

(j)

$$d^2 + 8d + 15 = ?$$

(k)

$$96a^3x + 48a^2x^2 + 6ax^3 = ?$$

(l)

$$x^3 - 4x = ?$$

(m)

$$4a^2 - b^2 + 2bc - c^2 = ?$$

(n)

$$3x^2y + 3xy - 60y = ?$$

(o)

$$81a^4 - 1 = ?$$

(p)

$$9 - 6x + x^2 = ?$$

(q)

$$a^2 - b^2 + 2b - 1 = ?$$

(r)

$$x^4 - 625 = ?$$

(s)

$$a^2 - 3ax - 3bx + ab = ?$$

Brüche

3. Aufgabe

Kürze die folgenden Brüche soweit wie möglich:

(a)

$$b_1 = \frac{13005}{5083} = ?$$

(b)

$$b_2 = \frac{15a - 6ab}{20c - 8bc} = ?$$

(c)

$$b_3 = \frac{a - x}{x - a} = ?$$

(d)

$$b_4 = \frac{(4a - 1)^2}{1 - 16a^2} = ?$$

(e)

$$b_5 = \frac{x^2 - 9x - 10}{x^2 - 1} = ?$$

(f)

$$b_6 = \frac{4x - x^3}{2x + x^2} = ?$$

4. Aufgabe

Berechne:

(a)

$$\frac{1}{x^2 - 2} + \frac{1}{2 - x} + \frac{1}{2 + x} = ?$$

(b)

$$\frac{a}{a^2 - 5a + 4} - \frac{a + 4}{a^2 + 3a - 4} = ?$$

(c)

$$x + \frac{x}{x - a} + a + \frac{a}{a - x} = ?$$

(d)

$$\frac{\frac{1}{x - \frac{1}{x}}}{\frac{x}{1 + \frac{x}{1}}} = ?$$

(e)

$$\frac{ax + y}{a + n} + \frac{nx - y}{a + n} = ?$$

(f)

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = ?$$

(g)

$$\frac{5x+3y}{3a} - \frac{2x+5y}{6b} - \frac{8x+6y}{6ab} = ?$$

(h)

$$\frac{2x+5}{4x-4} + \frac{5x-3}{6x-6} + \frac{2x+3}{2-2x} = ?$$

(i)

$$\frac{2n-3x}{3a+3} - \frac{5x+2n}{5b-5} + \frac{ax+bx}{ab-a+b-1} = ?$$

(j)

$$\frac{4a^2}{2a+b} + \frac{8ab^3}{(2a-b)(4a^2-b^2)} - \frac{b^2(2a+b)}{(2a-b)^2} = ?$$

(k)

$$\frac{\frac{1}{a^2-1} - 1}{1 - \frac{1}{a+1}} = ?$$

(l)

$$\frac{\frac{x^2}{1-x^2} + 1}{\frac{x}{1-x} + 1} : \frac{1}{1+x} = ?$$

(m)

$$\frac{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}{\frac{1}{x} - \frac{1}{y} - \frac{1}{z}} : \frac{\frac{2x^2}{x^2-z^2}}{\frac{x}{x+y} + \frac{x}{x-z}} = ?$$